

Test Cases of Hospital Management System

Test Case ID	Software Feature	Steps	Expected	Actual	Passed/Failed
TC1	Admin Login	1. Run the program. 2. Enter: admin 3. Enter: 123 -----Login----- Enter the username: admin Enter the password: 123	If credentials are correct hospital management system starts successfully. If credentials are incorrect login failed.	-----Login----- Enter the username: admin Enter the password: 123 Login Successful. -----Login----- Enter the username: hospital Enter the password: 123 Incorrect username or password.	Passed

TC2	Register Doctor	1. Run the program. 2. Choose the operation: <pre>Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 1</pre> 3. Enter Option: 1 4. Choose the operation: <pre>-----Doctor Management----- Choose the operation: 1 - Register 2 - View 3 - Update 4 - Delete Input: 1</pre> 5. Input: 1 6. Enter doctor's details: <pre>-----Register----- Enter the doctor's details: Enter the first name: John Enter the surname: Smith Enter the speciality: Nurse</pre>	If doctor's details are already in the system 'Name already exists' get display. If doctor's details are in the system then 'Doctor registered' get display.	<pre>-----Register----- Enter the doctor's details: Enter the first name: John Enter the surname: Smith Enter the speciality: Nurse Name already exists.</pre> <pre>-----Register----- Enter the doctor's details: Enter the first name: Bibek Enter the surname: Karki Enter the speciality: Dentist Doctor registered.</pre>	Passed
TC3	View doctor	1. Run the program. 2. Choose operation: <pre>-----Doctor Management----- Choose the operation: 1 - Register 2 - View 3 - Update 4 - Delete Input: 2</pre> 3. Input: 2	List of all the doctors in the system get display.	<pre>-----List of Doctors----- ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology 4 Dr. Bibek Karki Dentist</pre>	Passed

TC4	Update doctor	1. Run the program. 2. Choose operation: <pre> -----Doctor Management----- Choose the operation: 1 - Register 2 - View 3 - Update 4 - Delete Input: 3 </pre> 3. Input: 3 <pre> -----Update Doctor's Details----- ID Full name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology 4 Dr. Bibek Karki Dentist Enter the ID of the doctor: 4 </pre> 4. Enter doctor ID: 4 5. Choose field to update: 1/2/3 Choose: 1 <pre> Choose the field to be updated: 1 First name 2 Surname 3 Speciality Input: 1 </pre> 6. Enter new first name: Bishal <pre> Enter the new first name: Bishal </pre> Choose: 2 <pre> Choose the field to be updated: 1 First name 2 Surname 3 Speciality Input: 2 </pre> 8. Enter new surname: Thapa <pre> Enter the new surname: Thapa </pre> Choose : 3 <pre> Choose the field to be updated: 1 First name 2 Surname 3 Speciality Input: 3 </pre> 11. Enter new speciality: Surgeon <pre> Enter the new speciality: Surgeon </pre>	Doctor's new first name get updated in the system. Doctor's new surname get Updated in the system. Doctor's new speciality get updated in the system.	<pre> Enter the new first name: Bishal First name updated successfully </pre> <pre> Enter the new surname: Thapa Surname updated successfully </pre> <pre> Enter the new speciality: Surgeon Speciality updated successfully </pre> <pre> -----List of Doctors----- ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology 4 Dr. Bishal Thapa Surgeon </pre>	Passed
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TC5	Delete doctor	1. Run the program. 2. Choose operation: -----Doctor Management----- Choose the operation: 1 - Register 2 - View 3 - Update 4 - Delete Input: 4 3. Input: 4 -----Delete Doctor----- ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology 4 Dr. Bishal Thapa Surgeon Enter the ID of the doctor to be deleted: 4 4. Enter doctor ID to delete: 4	Entered doctor ID get deleted from the system.	Enter the ID of the doctor to be deleted: 4 Doctor deleted. -----List of Doctors----- ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology	Passed
TC6	Discharge Patients	1. Run the program. 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 2 3. Enter option: 2 -----View Patients----- ID Full Name Doctor's Full Name Age Mobile Postcode 1 Jane Smith Not Assigned 28 0782345678 10 2nd 2 Mike Jones Not Assigned 37 0755555234 12 3rd 3 Helen Jones Not Assigned 15 0732345678 13 4th Do you want to discharge a patient(Y/N): 4. Do you want to discharge patient(Y/N) Input: Y -----Discharge Patient----- Please enter the patient ID: 3 5. Enter patient ID: 3 Input: N Do you want to discharge a patient(Y/N): N	If 'Y' is entered patient get discharged from the system If 'N' is entered patient won't get discharge from the system.	-----Discharge Patient----- Please enter the patient ID: 3 Patient with ID 3 discharged successfully.	Passed

TC7	View Discharged Patients	1. Run the program. 2. Choose operation: <pre>Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 3</pre> 3. Enter option: 3	Discharged patients get display.	List of discharged patients display. <pre>-----Discharged Patients----- ID Full Name Doctor's Full Name Age Mobile Postcode 1 David Smith Not Assigned 15 07123456789 Q1 ABC</pre>	Passed
TC8	Assign doctors to patients.	1. Run the program. 2. Choose the operation: <pre>Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 4</pre> 3. Enter option: 4 <pre>-----Assign----- -----Patients----- ID Full Name Doctor's Full Name Age Mobile Postcode 1 Sara Smith Not Assigned 20 07812345678 RL 124 2 Mike Jones Not Assigned 37 07555551234 L2 100</pre> Please enter the patient ID: 1 4. Enter patient ID to assign to doctor: 1 <pre>----- ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology</pre> Please enter the doctor ID: 1 4. Enter the doctor ID: 1	Entered Patient ID get assign to entered doctor ID.	Display message showing that patient with selected ID get assign to selected doctor ID. <pre>-----Doctors Select----- Select the doctor that fits these symptoms: -----Symptoms----- 1. Hypertension 2. Ear Infection 3. Chest Discomfort ID Full Name Speciality 1 Dr. John Smith Internal Med. 2 Dr. Jone Smith Pediatrics 3 Dr. Jone Carlos Cardiology</pre> Please enter the doctor ID: 1 The patient Sara Smith is now assign to doctor Dr. John Smith.	Passed

TC9	Update admin details	1. Run the program. 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 5 3. Enter option: 5 4. Choose field : 1/2/3 Choose the field to be updated: 1 Username 2 Password 3 Address Input: 1 Input: 1 Enter new username: bibek Enter the new username: bibek Input: 2 Enter new password: 321 Enter the new password: 321 Enter the new password again: 321 Input: 3 Enter new address: KTM 7 Enter the new address: KTM 7	New username, password and adress get updated in the system	New user name, password and address get updated successfully in the system and store in admin.txt file. Enter the new username: bibek Username updated successfully. Enter the new password: 321 Enter the new password again: 321 Password updated successfully. Enter the new address: KTM 7 Address updated successfully. admin.txt 1 bibek 2 321 3 KTM 7 4	Passed
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TC 10	Patient Family Management	1. Run the program. 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 6 3. Enter option: 6	Patient with same family name are group accordingly.	Patient with same family name get group separately and display. Family Group----- Family: Smith <table><thead><tr><th>ID</th><th>Full Name</th><th>Doctor's Full Name</th><th>Age</th><th>Mobile</th><th>Postcode</th></tr></thead><tbody><tr><td>1</td><td>Sara Smith</td><td>Dr. John Smith</td><td>20</td><td>09012345678</td><td>BL 124</td></tr></tbody></table> Family: Jones <table><thead><tr><th>ID</th><th>Full Name</th><th>Doctor's Full Name</th><th>Age</th><th>Mobile</th><th>Postcode</th></tr></thead><tbody><tr><td>1</td><td>Wile Jones</td><td>Not Assigned</td><td>37</td><td>09555551234</td><td>LD 108</td></tr></tbody></table>	ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode	1	Sara Smith	Dr. John Smith	20	09012345678	BL 124	ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode	1	Wile Jones	Not Assigned	37	09555551234	LD 108	Passed
ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode																								
1	Sara Smith	Dr. John Smith	20	09012345678	BL 124																								
ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode																								
1	Wile Jones	Not Assigned	37	09555551234	LD 108																								

TC
11

Relocate Patients

1. Run the program.
2. Choose operation:
Choose the operation:
1- Register/view/update/delete doctor
2- Discharge patients
3- View discharged patient
4- Assign doctor to a patient
5- Update admin details
6- Patient Family Management
7- Relocate Patients
8- Schedule an appointment
9- Generate Management Report
10- Save Patient data to file
11- Load Patient data from file
12- Add Patient
13- View Report in diagram
14- Quit
Option: 7
3. Choose option: 7
-----Relocate Patient-----
-----Patients-----

ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode
1	Sara Smith	Dr. John Smith	28	07012345678	B1 234
2	Mike Jones	Not Assigned	37	07555551234	L2 345
3	David Smith	Not Assigned	15	07123456789	C1 ABC

Enter the ID of the patient to relocate: 1
4. Enter Patient ID to relocate:
1
-----Doctors Available-----

ID	Full Name	Speciality
1	Dr. John Smith	Internal Med.
2	Dr. Jone Smith	Pediatrics
3	Dr. Jone Carlos	Cardiology

Enter the ID of the new doctor: 2
5. Enter new doctor ID: 2

Patient with ID get relocate
from old doctor to new
doctor in the system.

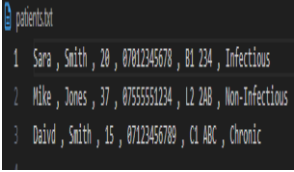
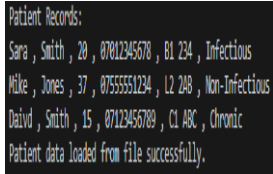
Patient get reassigned
from old doctor to new
doctor.

Enter the ID of the new doctor: 2
Patient Sara Smith has been relocated from Dr. John Smith to Dr. Jone Smith.

Passed

TC 12	Schedule an Appointment	1. Run the program. 2. Choose Operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 8 3. Choose Option: 8 -----Schedule Appointment----- -----Patients----- <table><tr><th>ID</th><th>Full Name</th><th>Doctor's Full Name</th><th>Age</th><th>Mobile</th><th>Postcode</th></tr><tr><td>1</td><td>Sara Smith</td><td>Not Assigned</td><td>20</td><td>0701245678</td><td>B1 124</td></tr><tr><td>2</td><td>Mike Jones</td><td>Not Assigned</td><td>37</td><td>0755551234</td><td>L2 24B</td></tr><tr><td>3</td><td>David Smith</td><td>Not Assigned</td><td>15</td><td>07123456789</td><td>C1 ABC</td></tr></table> Enter patient ID: 1 4. Enter Patient ID: 1 -----Doctors----- <table><tr><th>ID</th><th>Full Name</th><th>Speciality</th></tr><tr><td>1</td><td>Dr. John Smith</td><td>Internal Med.</td></tr><tr><td>2</td><td>Dr. Jone Smith</td><td>Pediatrics</td></tr><tr><td>3</td><td>Dr. Jone Carlos</td><td>Cardiology</td></tr></table> Enter doctor ID: 1 5. Enter Doctor ID: 1 Enter appointment date (YYYY-MM-DD): 2025-3-7 Enter appointment time (H:M, 12-hour format(H:M AM/PM)): 3:00 PM 6. Enter Appointment Date: 7. Enter Appointment Time:	ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode	1	Sara Smith	Not Assigned	20	0701245678	B1 124	2	Mike Jones	Not Assigned	37	0755551234	L2 24B	3	David Smith	Not Assigned	15	07123456789	C1 ABC	ID	Full Name	Speciality	1	Dr. John Smith	Internal Med.	2	Dr. Jone Smith	Pediatrics	3	Dr. Jone Carlos	Cardiology	Entered patient ID with appointment date and time get appointed to respective doctor.	Display message showing patient is appointed to doctor at respective date and time. Enter appointment date (YYYY-MM-DD): 2025-3-7 Enter appointment time (H:M, 12-hour format(H:M AM/PM)): 3:00 PM Appointment scheduled for Sara Smith with Dr. John Smith on 2025-3-7 at 3:00 PM.	Passed
ID	Full Name	Doctor's Full Name	Age	Mobile	Postcode																																				
1	Sara Smith	Not Assigned	20	0701245678	B1 124																																				
2	Mike Jones	Not Assigned	37	0755551234	L2 24B																																				
3	David Smith	Not Assigned	15	07123456789	C1 ABC																																				
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1	Dr. John Smith	Internal Med.																																							
2	Dr. Jone Smith	Pediatrics																																							
3	Dr. Jone Carlos	Cardiology																																							

TC 13	Generate Management Report	1. Run the program. 2. Choose operation: <pre>Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 9</pre> 3. Enter option: 9 <pre>----Management Report---- 1- Total Doctors 2- Patient per doctor 3- Appointment per month per doctor 4- Patient by illness type 5- Return to Main Menu Enter option: 1</pre> 4. Enter option: 1 5. Enter option: 2 6. Enter option: 3 7. Enter option: 4 8. Enter option: 5	If option '1' is entered then total number of doctors get display. If option '2' is entered then total number of patient per doctor get display. If option '3' is entered then number of appointment per month per doctor get display. If option '4' is entered then patient by illness type get display. If option '5' is entered then return to main menu.	1.Total Number of doctor display <pre>1. Total Doctors: 3</pre> 2.Number of patient per doctor display <pre>Patients per doctor: ----- Doctor Name No. of Patients ----- Dr. John Smith 0 Dr. Jone Smith 2 Dr. Jone Carlos 1 -----</pre> 3.Number of appointment per month per doctor get display. <pre>3. Appointments per month per doctor: Appointments for Dr. John Smith: ----- Month No. of Appointments ----- 3 1 5 1 -----</pre> <pre>Appointments for Dr. Jone Smith: ----- Month No. of Appointments ----- 4 1 -----</pre> <pre>Appointments for Dr. Jone Carlos: ----- Month No. of Appointments ----- 9 1 -----</pre> Patients by Illness type get display. <pre>4. Patients by Illness Type: ----- Illness Category No. of Patients ----- Infectious 1 Non-Infectious 1 Chronic 1 -----</pre>	Passed
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TC 14	Save Patient Data to File	1.Run the program 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 10 3. Enter option: 10	All the patients data store in the file patients.txt.	Patients data store in file patients.txt successfully. 	Passed
TC 15	Load Patient Data from File	1.Run the program 2.Choose operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 11	Patients records get load from the file to system and get display in terminal.	Patients records get display in the terminal . 	Passed

TC 16	Add Patients	1. Run the program. 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit - Option: 12 3. Enter option: 12 Enter first name of patient: Bibek Enter last name of patient: Karki Enter age of patient: 21 Enter phone number of patient: 873672762 Enter postcode of patient: HDT34 Enter illness type of patient: Chronic 4. Enter first name: 5. Enter surname: 6. Enter age: 7. Enter phone number: 8. Enter postcode: 9. Enter illness type:	‘Patients added and saved successfully’ message display in terminal and patients data get added to the file patients.txt	Enter first name of patient: Bibek Enter last name of patient: Karki Enter age of patient: 21 Enter phone number of patient: 873672762 Enter postcode of patient: HDT34 Enter illness type of patient: Chronic Patient added and saved successfully. Patients data get added in the file ‘patients.txt’ <pre> patients.txt 1 Sara, Smith, 20, 07012345678, B1 234, Infectious 2 Mike, Jones, 37, 07555551234, L2 24B, Non-Infectious 3 David, Smith, 15, 07123456789, C1 ABC, Chronic 4 Bibek, Karki, 21, 873672762, HDT34, Chronic 5 </pre>	Passed
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TC 17	Quit	1.Run the program 2. Choose operation: Choose the operation: 1- Register/view/update/delete doctor 2- Discharge patients 3- View discharged patient 4- Assign doctor to a patient 5- Update admin details 6- Patient Family Management 7- Relocate Patients 8- Schedule an appointment 9- Generate Management Report 10- Save Patient data to file 11- Load Patient data from file 12- Add Patient 13- View Report in diagram 14- Quit Option: 14 3. Enter option: 14	'Goodbye' message display in terminal	-----Goodbye-----	Passed
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